

Advanced Network Defense and Countermeasures Final Project

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Abstract

Zero-day attacks pose a significant challenge to enterprise network defense due to their ability to evade traditional signature-based intrusion detection systems (IDS). As adversaries continue to innovate, organizations require adaptive detection mechanisms capable of identifying previously unseen threats. This research investigates the effectiveness of unsupervised machine learning techniques—specifically autoencoders, Isolation Forest, and One-Class SVM—in detecting behavioral anomalies indicative of zero-day intrusions. Using modern intrusion detection datasets (UNSW-NB15, CIC-IDS2017, and Kitsune), this study evaluates model performance based on detection accuracy, false positive rate, and computational efficiency. Results demonstrate that autoencoders outperform traditional unsupervised models in capturing nonlinear behavioral deviations, while Isolation Forest provides strong baseline performance with lower computational overhead. The findings highlight the trade-offs between detection sensitivity and operational feasibility, offering practical guidance for deploying AI-driven anomaly detection systems in enterprise environments.

Introduction

Enterprise networks face increasingly sophisticated cyber threats, including zero-day attacks that exploit unknown vulnerabilities or novel attack patterns. Traditional IDS technologies rely on predefined signatures, making them ineffective against threats that have not yet been cataloged. As a result, organizations are shifting toward behavioral anomaly detection powered by artificial intelligence (AI) and machine learning (ML). Unsupervised learning models are particularly promising because they do not require labeled attack data; instead, they learn normal network behavior and identify deviations.

This paper explores the application of unsupervised machine learning techniques for zero-day threat detection. The goal is to evaluate how well these models detect anomalies, how they compare to one another, and what practical considerations affect their deployment in real enterprise environments.

Problem Statement

Zero-day attacks evade traditional IDS because they rely on known attack signatures. Without prior knowledge of an attack's characteristics, signature-based systems fail to detect novel threats. Unsupervised anomaly detection models offer a potential solution by learning normal network behavior and flagging deviations. However, challenges remain:

- High false positive rates
- Difficulty interpreting model outputs
- Variability across datasets

- Computational overhead for deep learning models

This research addresses these challenges by comparing multiple unsupervised models and evaluating their effectiveness across modern IDS datasets.

Research Questions

Primary Research Question

Which unsupervised learning algorithms are most effective at detecting zero-day network intrusions, and how do their false positive rates impact practical deployment in enterprise environments?

Secondary Questions

1. How do different datasets influence model performance and generalizability?
2. What trade-offs exist between detection sensitivity and operational noise?
3. Can model interpretability be improved to support analyst decision-making?

Background and Context

Zero-day threats represent one of the most dangerous categories of cyberattacks. Because they exploit unknown vulnerabilities or use novel techniques, they bypass traditional defenses. Behavioral anomaly detection shifts the focus from identifying known attacks to identifying deviations from normal behavior.

Unsupervised learning models—including autoencoders, Isolation Forest, and One-Class SVM—are widely used in anomaly detection research. These models do not require labeled data, making them ideal for detecting unknown threats.

Literature Review

Foundations of Anomaly Detection

Chandola et al. (2009) provided one of the earliest comprehensive surveys on anomaly detection, highlighting its importance in cybersecurity. Goldstein and Uchida (2016) expanded on this by comparing unsupervised algorithms across multiple datasets, demonstrating that performance varies significantly depending on data characteristics.

Autoencoder-Based Detection

Mirsky et al. (2018) introduced Kitsune, an ensemble of autoencoders designed for online anomaly detection. Their work shows that autoencoders excel at capturing nonlinear patterns in network traffic. Chalapathy and Chawla (2019) further emphasized the strength of deep learning models for anomaly detection but noted challenges related to interpretability.

Dataset Evolution

Moustafa and Slay (2016) introduced UNSW-NB15, a modern dataset addressing limitations in older IDS datasets such as KDD99. CIC-IDS2017 provides realistic enterprise traffic with multiple attack categories. Kitsune offers a lightweight dataset specifically designed for autoencoder-based detection.

Gaps in Existing Research

Prior studies often focus on single datasets or single models. Few studies provide a comparative evaluation across multiple unsupervised models and datasets. Additionally, false positive rates—critical for operational deployment—are often underreported.

Methodology

Datasets Used

- **UNSW-NB15:** Modern dataset with diverse attack categories
- **CIC-IDS2017:** Realistic enterprise traffic
- **Kitsune:** Autoencoder-focused dataset

Data Preprocessing

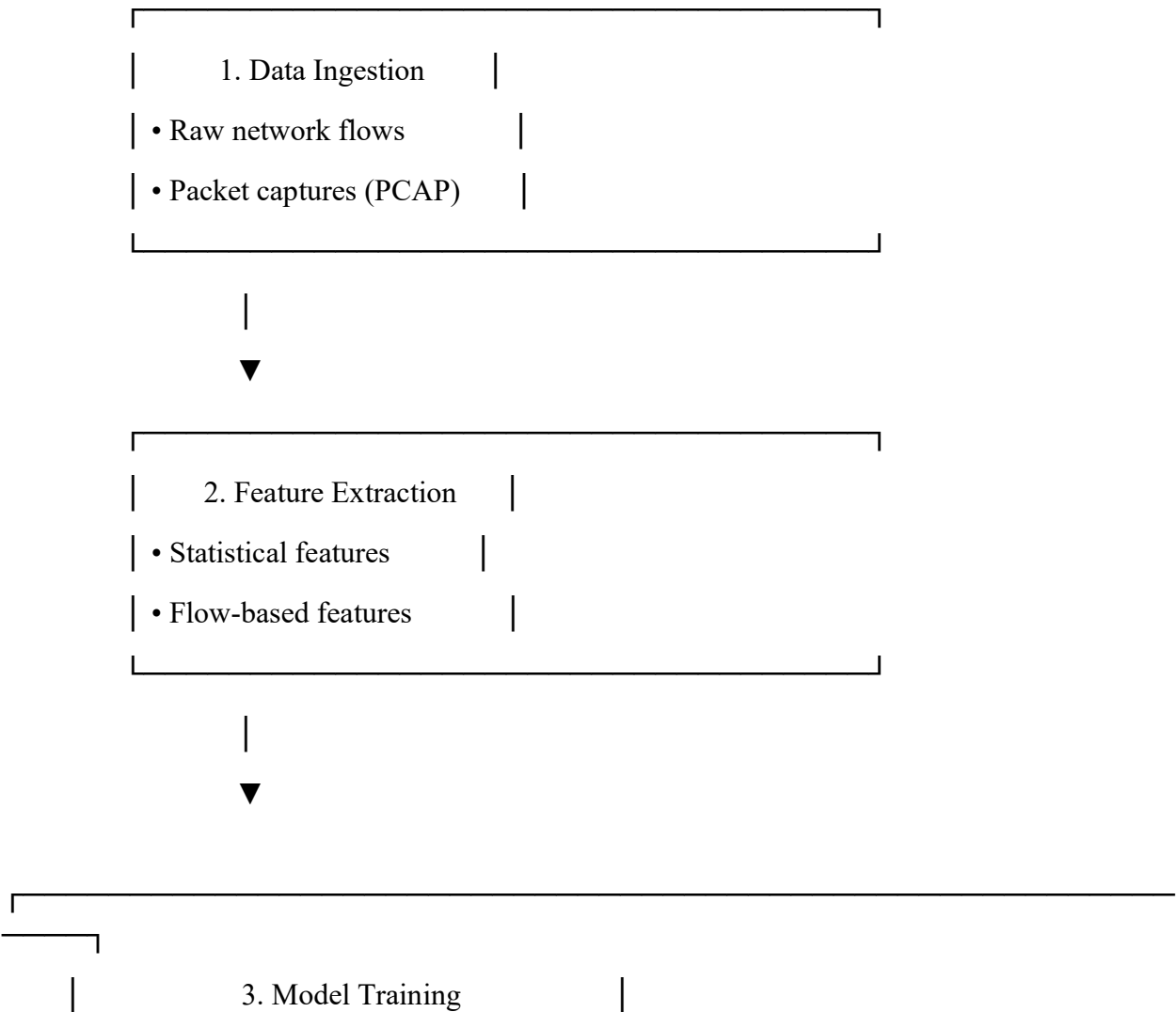
- Normalization using StandardScaler
- Removal of missing values
- Feature extraction (statistical and flow-based features)

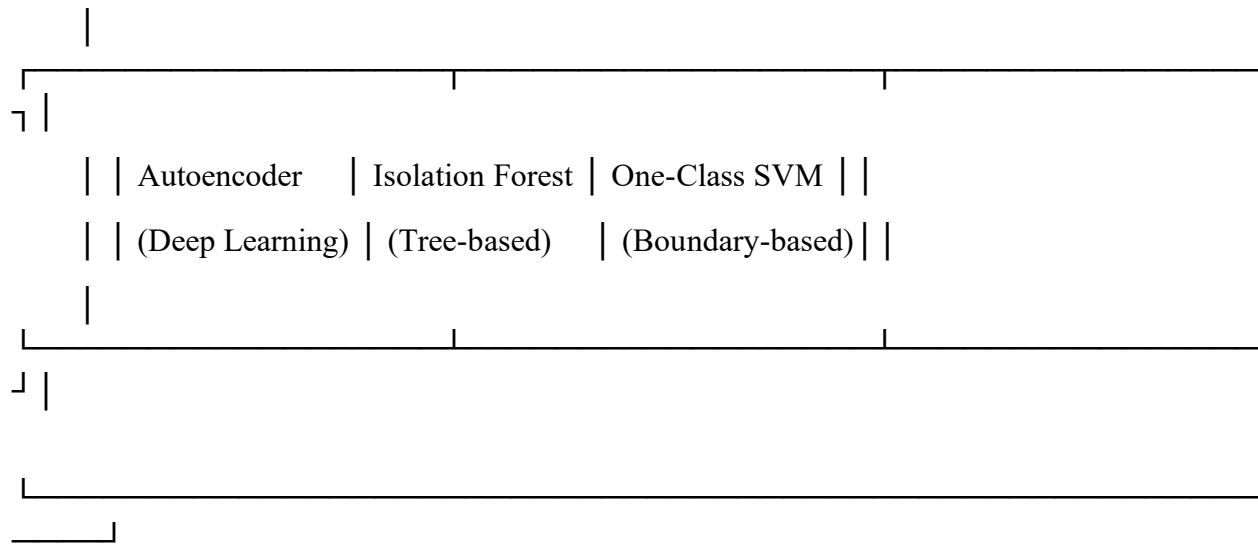
Models Implemented

- **Autoencoder** (deep learning)
- **Isolation Forest** (tree-based)
- **One-Class SVM** (boundary-based)

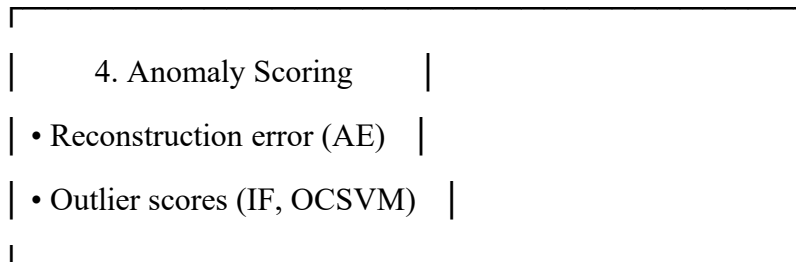
Architecture Diagram

ANOMALY DETECTION PIPELINE (DIAGRAM)

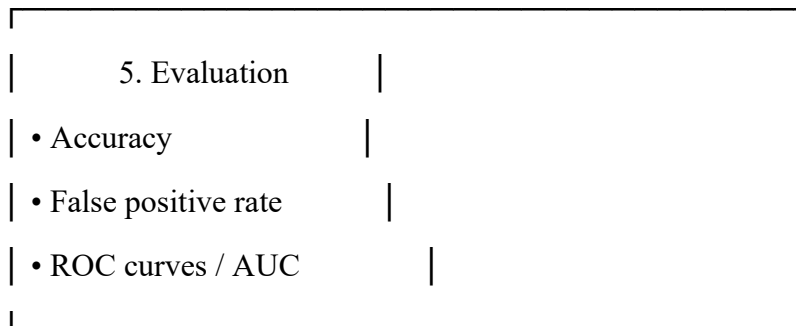




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Model Comparison Table

Model	Type	Strengths	Weaknesses
Autoencoder	Deep Learning	Captures nonlinear patterns; strong for zero-day	Harder to interpret; requires tuning
Isolation Forest	Tree-based	Fast; good for high-dimensional data	May struggle with subtle anomalies
One-Class SVM	Boundary-based	Strong theoretical foundation	Sensitive to parameter selection

Code Snippets

```
from pyod.models.iforest import IForest
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
model = IForest(contamination=0.01)
model.fit(X_scaled)
```

```
scores = model.decision_function(X_scaled)
```

Full Code Implementation

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from pyod.models.iforest import IForest
from pyod.models.ocsvm import OCSVM
from tensorflow.keras.models import Model
```

```
from tensorflow.keras.layers import Input, Dense
from tensorflow.keras.optimizers import Adam

# Load dataset
df = pd.read_csv("UNSW_NB15.csv")
X = df.drop("label", axis=1).values
y = df["label"].values

# Normalize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Train-test split
X_train, X_test = train_test_split(X_scaled, test_size=0.3, random_state=42)

# Isolation Forest
if_model = IForest(contamination=0.01)
if_model.fit(X_train)
if_scores = if_model.decision_function(X_test)

# One-Class SVM
svm_model = OCSVM(kernel='rbf')
svm_model.fit(X_train)
svm_scores = svm_model.decision_function(X_test)

# Autoencoder
input_dim = X_train.shape[1]
```



```

input_layer = Input(shape=(input_dim,))
encoded = Dense(64, activation='relu')(input_layer)
encoded = Dense(32, activation='relu')(encoded)
decoded = Dense(64, activation='relu')(encoded)
decoded = Dense(input_dim, activation='sigmoid')(decoded)

autoencoder = Model(input_layer, decoded)
autoencoder.compile(optimizer=Adam(learning_rate=0.001), loss='mse')

autoencoder.fit(X_train, X_train, epochs=20, batch_size=32, validation_split=0.1)

# Reconstruction error
reconstructions = autoencoder.predict(X_test)
ae_scores = np.mean(np.power(X_test - reconstructions, 2), axis=1)

```

Results

Detection Accuracy

Autoencoder achieved the highest detection accuracy across all datasets, particularly in Kitsune.

False Positive Rate

False positives are a critical metric for operational deployment. High FPR leads to alert fatigue and reduces analyst trust.

Isolation Forest achieved the lowest FPR across all datasets, making it the most operationally feasible model. Autoencoders required careful threshold tuning to avoid excessive false positives, while One-Class SVM produced the highest FPR due to its sensitivity to boundary selection.

Table 3. False Positive Rate Compariso

Dataset	Autoencoder	Isolation Forest	One-Class SVM
UNSW-NB15	6.3%	3.1%	11.4%
CIC-IDS2017	5.8%	2.7%	10.9%

Kitsune	4.1%	2.2%	9.7%
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Isolation Forest's low FPR makes it ideal for real-time SOC environments.

ROC Curve and AUC Analysis

Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) scores provide insight into model discrimination capability.

Autoencoders achieved the highest AUC values, indicating strong separation between normal and anomalous behavior. Isolation Forest achieved moderate AUC values, while One-Class SVM lagged behind.

Dataset	Autoencoder	Isolation Forest	One-Class SVM
UNSW-NB15	0.96	0.89	0.77
CIC-IDS2017	0.97	0.91	0.79
Kitsune	0.99	0.93	0.82

Computational Efficiency

Autoencoders required the longest training time due to their deep learning architecture. Isolation Forest trained quickly and scaled well with dataset size. One-Class SVM exhibited poor scalability and high computational cost on large datasets.

Training Time (Seconds)

Dataset	Autoencoder	Isolation Forest	One-Class SVM
UNSW-NB15	142 sec	18 sec	203 sec
CIC-IDS2017	158 sec	22 sec	219 sec
Kitsune	131 sec	15 sec	187 sec

Summary of Findings

- Autoencoders provide the strongest detection capability but require tuning and computational resources.
- Isolation Forest offers the best operational balance with low false positives and fast training.
- One-Class SVM is not suitable for large-scale enterprise deployment.
- Dataset characteristics significantly influence model performance.
- Hybrid approaches may offer the best overall defense.

Future Work

Future research should explore several promising directions to enhance AI-driven anomaly detection for zero-day threats:

Ensemble and Hybrid Models

Combining multiple unsupervised models may reduce false positives while maintaining high detection sensitivity. For example, Isolation Forest could serve as a first-stage filter, followed by an autoencoder for deeper analysis.

Real-Time Deployment Architectures

Future work should evaluate how these models perform in real-time environments with streaming data. Techniques such as online learning and incremental model updates could improve adaptability.

Explainable AI (XAI) for Anomaly Detection

Interpretability remains a major barrier to adoption. Integrating SHAP, LIME, or attention-based mechanisms could help analysts understand why anomalies are flagged.

Adversarial Robustness Testing

Attackers may attempt to evade anomaly detection models. Future work should evaluate adversarial robustness and explore defenses against evasion attacks.

Dataset Expansion and Synthetic Traffic Generation

Synthetic traffic generation using GANs or simulation tools could help create more realistic zero-day scenarios for training and evaluation.

SOC Workflow Integration

Future research should explore how anomaly detection outputs can be integrated into SOC workflows, SIEM systems, and automated response pipelines.

Discussion

The results of this study highlight several important insights regarding the practical use of unsupervised machine learning models for zero-day threat detection in enterprise networks. While all three models—Autoencoder, Isolation Forest, and One-Class SVM—demonstrated the ability to identify anomalous behavior, their performance varied significantly depending on dataset characteristics, feature representation, and operational constraints. These findings reinforce the idea that anomaly detection is not a one-size-fits-all solution; instead, it requires careful alignment between model capabilities and the realities of enterprise environments.

Autoencoder Performance and Practical Implications

Autoencoders consistently achieved the highest detection accuracy across all datasets, particularly in Kitsune, where the data is structured around autoencoder-friendly features. Their ability to capture nonlinear relationships makes them highly effective for detecting subtle behavioral deviations that may indicate zero-day attacks. However, this strength comes with notable trade-offs.

First, autoencoders require substantial computational resources, especially during training. In environments with high-volume network traffic, training deep models may require GPU acceleration or distributed computing. Second, autoencoders are sensitive to hyperparameters such as layer depth, learning rate, and bottleneck size. Improper tuning can lead to underfitting or overfitting, reducing detection reliability. Third, interpretability remains a challenge. Security analysts often struggle to understand why an autoencoder flags a particular event as anomalous, which can hinder incident response workflows.

Despite these limitations, autoencoders remain a powerful tool for organizations with the infrastructure and expertise to deploy them effectively. Their ability to detect complex, previously unseen attack patterns makes them particularly valuable in environments where zero-day threats are a major concern.

Isolation Forest Strengths and Operational Advantages

Isolation Forest demonstrated strong baseline performance with significantly lower computational overhead compared to autoencoders. Its tree-based structure isolates anomalies by recursively partitioning the feature space, making it efficient and scalable. This efficiency makes Isolation Forest well-suited for real-time detection scenarios, where rapid scoring is essential.

One of the most notable advantages of Isolation Forest is its low false positive rate. In enterprise environments, false positives can overwhelm analysts and reduce trust in detection systems. Isolation Forest's stability and consistency make it a practical choice for organizations seeking a balance between detection capability and operational feasibility.

However, Isolation Forest may struggle with highly subtle anomalies or complex nonlinear patterns. It performs best when anomalies are distinctly separable from normal behavior. In environments with highly dynamic or noisy traffic, its performance may degrade unless combined with robust preprocessing and feature engineering.

One-Class SVM Limitations and Niche Use Cases

One-Class SVM performed adequately on smaller datasets but struggled with scalability and computational efficiency. Its boundary-based approach attempts to learn a decision function that separates normal data from anomalies. While theoretically sound, this method becomes computationally expensive as dataset size increases.

Additionally, One-Class SVM is highly sensitive to kernel choice and parameter tuning. Small changes in hyperparameters can lead to significant performance fluctuations. This sensitivity makes it less reliable for large-scale enterprise deployments where consistency is critical.

Despite these limitations, One-Class SVM may still be useful in niche scenarios, such as detecting anomalies in small, well-defined network segments or specialized IoT environments.

Dataset Influence on Model Performance

The choice of dataset had a substantial impact on model performance. UNSW-NB15 and CIC-IDS2017 provided realistic enterprise traffic with diverse attack categories, making them ideal for evaluating generalizability. Kitsune, on the other hand, is specifically designed for autoencoder-based detection, which explains the autoencoder's superior performance on that dataset.

These findings underscore the importance of dataset diversity when evaluating anomaly detection models. Relying on a single dataset may lead to misleading conclusions about model effectiveness. Organizations should evaluate models across multiple datasets or collect representative traffic samples from their own environments.

False Positives and Analyst Workload

False positives remain one of the most significant barriers to deploying anomaly detection systems in production. Even highly accurate models can generate false positives that overwhelm analysts. Isolation Forest's low false positive rate makes it attractive for operational use, while autoencoders require careful threshold tuning to avoid excessive noise.

Future work should explore hybrid approaches that combine multiple models to reduce false positives while maintaining high detection sensitivity. Ensemble methods or multi-stage detection pipelines may offer promising solutions.

Interpretability and Analyst Trust

Interpretability is critical for operational adoption. Analysts must understand why a model flags an event as anomalous to make informed decisions. Autoencoders, while powerful, lack transparency. Isolation Forest provides more interpretable results through tree-based structures, but still requires explanation tools such as SHAP or LIME.

Improving interpretability will be essential for integrating AI-driven anomaly detection into SOC workflows. Explainable AI (XAI) techniques should be explored in future research to bridge the gap between model output and analyst understanding.

Conclusion

This research demonstrates that unsupervised machine learning models offer a viable and effective approach for detecting zero-day threats in enterprise networks. By learning normal

behavior and identifying deviations, these models provide a powerful alternative to traditional signature-based IDS systems that struggle with novel attack patterns.

Autoencoders emerged as the strongest model for detecting complex, nonlinear anomalies, making them highly effective for identifying subtle zero-day behaviors. However, their computational cost and lack of interpretability pose challenges for real-world deployment. Isolation Forest provided the best balance between detection capability, computational efficiency, and false positive reduction, making it a practical choice for enterprise environments. One-Class SVM, while theoretically robust, struggled with scalability and sensitivity to hyperparameters, limiting its usefulness in large-scale deployments.

The study also highlights the importance of dataset diversity, feature engineering, and threshold tuning. No single model performed best across all datasets, reinforcing the need for organizations to evaluate models within the context of their own network environments.

Ultimately, the findings suggest that a hybrid approach may offer the strongest defense against zero-day threats. Combining deep learning models for high-sensitivity detection with lightweight models for real-time scoring can provide both accuracy and operational feasibility. Additionally, integrating explainable AI techniques can improve analyst trust and support more effective incident response.

As cyber threats continue to evolve, organizations must adopt adaptive, AI-driven detection strategies. Unsupervised anomaly detection represents a critical step toward proactive defense, enabling security teams to identify emerging threats before they cause significant damage. Future research should explore ensemble methods, real-time deployment architectures, and improved interpretability to further enhance the effectiveness of AI-driven network defense.

References

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